

Black holes & gravitational waves – Exam 2026

Guidelines

- This is a take-home exam taking place from *9:00am to 5:00pm on Monday, June 15, 2026*. It will be graded. It will count for 60% of the total grade. Please hand-in your answers **before 5pm**.
- Answers are handed digitally through e-mail. **Please send your answer to question 1 and 2 in separate files to:**
maarten.meent@nbi.ku.dk & jose.ezquiaga@nbi.ku.dk.
Please name the files:
LastName_FirstName_exam_2026_Q1.pdf
LastName_FirstName_exam_2026_Q2.pdf
- You must do the exam **individually**. It is **not allowed** to use AI tools like ChatGPT or Claude. We need to evaluate what *you* have learned during the course.
- You are **allowed** to use computer algebra software (e.g. Mathematica) and coding in general (e.g. Jupyter notebook or Python) to aid with your computations, if you write the code yourself. However, if you do, please include your notebooks with your answers.
- Please make sure that your answers (and any supplemental material such as code) are comprehensible!

Question 1: Bob and Bill's black hole adventure

Bob and Bill are playing near a Kerr black hole with spin $a = 0.95M$, and have gotten themselves in a bit of bind.

In a desperate attempt to save themselves they push away from each other at a spacetime position with Boyer-Lindquist coordinates

$$\begin{aligned}t &= 0, \\r &= 1.5M, \\ \cos \theta &= 0.5 \\ \phi &= 0.\end{aligned}$$

Bob heads-off with 4-velocity

$$\begin{aligned}\frac{dt}{d\tau} &= 9.05, \\ \frac{dr}{d\tau} &= 0, \\ \frac{d\theta}{d\tau} &= 0 \\ \frac{d\phi}{d\tau} &= 3.47/M\end{aligned}$$

and Bill with 4-velocity

$$\begin{aligned}\frac{dt}{d\tau} &= 5.99, \\ \frac{dr}{d\tau} &= 0, \\ \frac{d\theta}{d\tau} &= 0 \\ \frac{d\phi}{d\tau} &= 1.38/M\end{aligned}$$

- a. Calculate the constants of motion μ , $\mathcal{E}$, $\mathcal{L}$, $\mathcal{Q}$ for Bob and Bill. (Round to 3 significant digits.)
- b. How many real roots does the radial potential P_r have for Bob? And how many for Bill?
- c. What kind of geodesics will Bob and Bill follow.
- d. Draw the Penrose diagram for the maximally extended Kerr spacetime, and sketch the trajectories of Bob and Bill.
- e. Will Bob and Bill ever see each other again?

Question 2: the universal distribution of gravitational-wave detections

As we have seen in the course, the *optimal* signal-to-noise ratio (SNR) ρ of a gravitational-wave (GW) signal $h(t)$ is obtained when one filters the data with a template that is equal to the signal itself

$$\rho^2 = (h|h),$$

where the noise-weighted inner product is defined by

$$(a|b) = 4\text{Re} \int_0^\infty \frac{\tilde{a}^*(f)b(f)}{S_n(f)} df.$$

In the wave frame, the GW is characterized by its two polarizations $h_{+,\times}(t)$. In the detector frame, one needs to project using the *antenna pattern functions*:

$$h(t) = F_+(\theta, \phi, \psi)h_+(t) + F_\times(\theta, \phi, \psi)h_\times(t),$$

which for L-shaped detectors depend on the sky localization $\{\theta, \phi\}$ and the polarization angle ψ as

$$\begin{aligned} F_+(\theta, \phi, \psi) &= \frac{1}{2} \left(1 + \cos^2 \theta \right) \cos 2\phi \cos 2\psi - \cos \theta \sin 2\phi \sin 2\psi, \\ F_\times(\theta, \phi, \psi) &= \frac{1}{2} \left(1 + \cos^2 \theta \right) \cos 2\phi \sin 2\psi + \cos \theta \sin 2\phi \cos 2\psi. \end{aligned}$$

If we consider a network of *identical* detectors, $k = 1, \dots, N$, the network SNR is simply

$$\rho_N^2 = \sum_k \rho_k^2,$$

where the SNR in each detector ρ_k , varies because of the different locations that change the antenna pattern functions $F_{+, \times, k}$, thus changing the projected strain

$$h_k(t) = F_{+, k}(\theta, \phi, \psi) h_+(t) + F_{\times, k}(\theta, \phi, \psi) h_\times(t).$$

1. Show that when taking an average over the polarization angle, the angular dependence of the SNR factors out; specifically:

$$\langle \rho^2 \rangle = 2P(\theta, \phi) \int_0^\infty \frac{|\tilde{H}(f)|^2}{S_n(f)} df,$$

where we have defined the *antenna power pattern*

$$P(\theta, \phi) \equiv F_+(\theta, \phi, \psi)^2 + F_\times(\theta, \phi, \psi)^2$$

and

$$|\tilde{H}(f)|^2 = |\tilde{h}_+(f)|^2 + |\tilde{h}_\times(f)|^2.$$

If we define the *visibility distance* of the source D_V , as the distance at which the polarization-averaged SNR for an overhead signal ($\theta = 0$ or π) is 1. How does $\langle \rho^2 \rangle$ depend on D_V ?

2. Generalize the previous point to a network of identical detectors, i.e. compute how $\langle \rho_N^2 \rangle$ depends on the *network antenna pattern power*

$$P_N(\theta, \phi) \equiv \sum_k \left(F_{+, k}(\theta, \phi, \psi)^2 + F_{\times, k}(\theta, \phi, \psi)^2 \right)$$

and then express it in terms of the visibility distance D_V .

3. Assuming an *Euclidean universe*, the *detection volume* is determined by

$$V_N = \int d\Omega \int_0^{R_N} r^2 dr,$$

where R_N is the network *range distance*

$$R_N = \frac{D_V}{\rho_{N, \text{th}}} P_N(\theta, \phi)^{1/2},$$

$\rho_{N,\text{th}}$ is the *network threshold SNR*, and $d\Omega$ the differential solid angle. Show that because the number of detections above a network threshold scales with the volume, the SNR follows the *universal* distribution

$$p(\rho_N) = 3 \frac{\rho_{N,\text{th}}^3}{\rho_N^4}, \quad \text{for } \rho_N > \rho_{N,\text{th}}. \quad (0.0.1)$$

For a given threshold, $\rho_{N,\text{th}}$, what is the average network SNR? And the median?

4. If we define the *loudest* GW in a catalog of Z *independent* events as the one with maximum SNR, $\rho_{\text{max}} = \max\{\rho_1, \dots, \rho_Z\}$, the probability of ρ_{max} being smaller than a given value is given by

$$\begin{aligned} P(\rho_{\text{max}} < \rho) &= P(\rho_1 < \rho; \rho_2 < \rho; \dots; \rho_Z < \rho) \\ &= P(\rho_1 < \rho) P(\rho_2 < \rho) \dots P(\rho_Z < \rho) \\ &= (F_\rho)^Z, \end{aligned}$$

where F_ρ is the cumulative distribution of the universal SNR distribution from the threshold SNR to ρ . If the network threshold SNR is 12, what is the probability that after 1000 events we observe at least one with SNR larger than 100?

5. Focusing on the leading quadrupolar radiation, how does the detection volume depend on the inclination of the binary ι ? How does the *average* number of detections over uniformly distributed inclinations compare to the case in which all the binaries are face-on ($\iota = 0$)? How does the probability distribution of detected binaries compare to the intrinsic one?